UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

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|---|---|
| COURSE | PRE CALCULUS |
| COURSE CODE | MAT133 |
| EXAMINATION | OCTOBER 2026 |
| TIME | 2 HOURS |

INSTRUCTIONS TO MARKERS

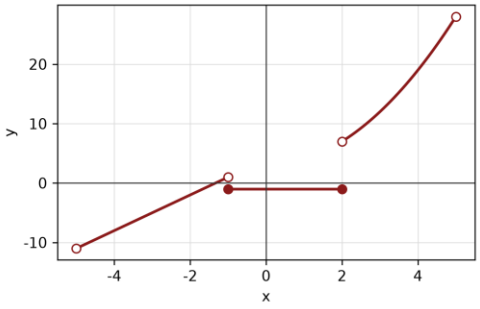
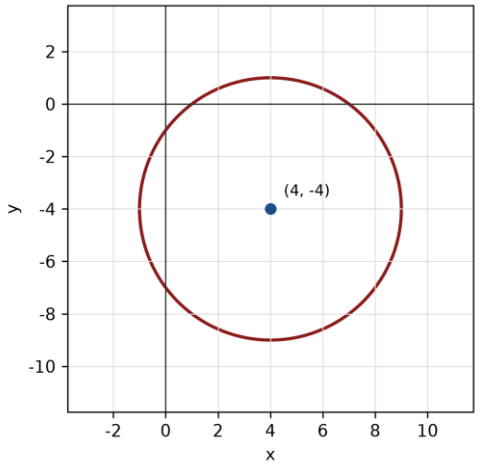
1. This marking scheme accompanies the MAT133 mock paper (October 2026).
2. Answers and marking points are for local study use only.

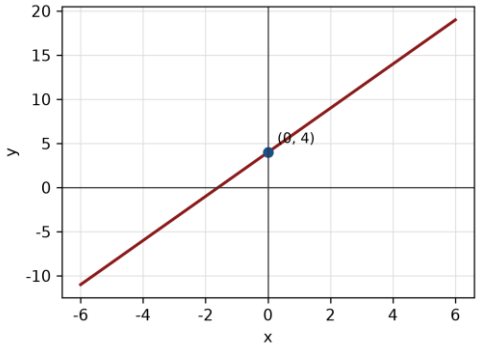
FOR MARKERS' USE ONLY

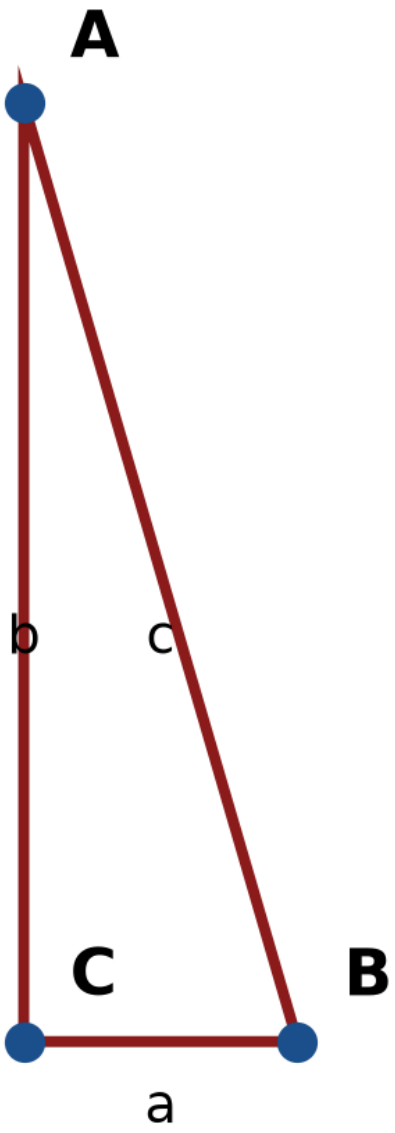
*This marking scheme consists of 5 printed pages*

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| NO | SOLUTION | | MARK |
|---|---|---|---|
| 1 a) | $-x-1>-3$<br>$-x>-2$<br>$(-\infty, 2)$<br>  | M1<br><br>A1 | 3 m |
| 1 b) | $\lvert 4x+2\rvert - 3 \geq 0$<br>$-4x-2 >= 3$ or $-4x-2 <= -3$<br>$\left(-\infty, -\frac{5}{4}\right] \cup \left[\frac{1}{4}, \infty\right)$<br>  | M1<br>A1<br>A1 | 5 m |
| 1 c) | $\lvert 5x-2\rvert = 9$<br>$2-5x=9$<br>$2-5x=-9$ | M1<br>A1<br>A1 | 4 m |
| 2 a) | $(f+g)(x)=-3$<br>$(f-g)(x)=2x+5$<br>$(fg)(x)=-x^2-5x-4$ | M1<br>A1<br>B1 | 4 m |

| NO | SOLUTION | | MARK |
|---|---|---|---|
| 2 b) | $f(x) = 3x + 4, \quad x < -1$<br>$f(x) = -1, \quad -1 \le x \le 2$<br>$f(x) = x^2 + 3, \quad x > 2$<br>$f(-2) = -2, \quad f(0) = -1, \quad f(3) = 12$<br>  <br> | B1<br><br><br><br>A1 | 5 m |
| 2 c) | $x^2 - 8x + y^2 + 8y = -7$<br>$(x-4)^2 + (y+4)^2 - 32 = -7$<br>$(x-4)^2 + (y+4)^2 = 25$<br>  <br> | M1 M1<br><br>A1<br><br>A1 | 5 m |
| 3 a) | $(-5 - 6i) \times (4 + 6i)$<br>$16 - 54i$ | M1 A1<br><br>A1 | 4 m |
| 3 b) | $\lvert z \rvert = \sqrt{4^2 + 3^2} = 5$<br>$\tan\theta = \frac{3}{4}, \quad \theta = 36.9^\circ$ | M1 A1 M1<br><br>A1 | 5 m |

| NO | SOLUTION | | MARK |
|---|---|---|---|
| 3 c) | $2x^3 - 2x^2 - 24x$ | M1 | 5 m |
| | $f(0) = 0, \quad f(-3) = 0, \quad f(4) = 0$ | M1 | |
| | $2x(x-4)(x+3)$ | A1 | |
| 4 a) | $2x + 3y + z = -7$ | M1 | 5 m |
| | $-x - 3y - 2z = 2$ | A1 | |
| | $-2x + 2y - z = 2$ | | |
| | $x = -3, \quad y = -1, \quad z = 2$ | A1 | |
| 4 b) | $4x - 1 = x^2 - 1$ | M1 | 5 m |
| | $x^2 - 4x = 0$ | A1 | |
| | $x(x-4) = 0$ | | |
| | $x = 4 \text{ or } x = 0$ | | |
| | $(4, 15), \quad (0, -1)$ | A1 | |
| 4 c) | $-5x + 2y - 7 = 0$ | M1 | 5 m |
| | $y = \frac{5x}{2} + \frac{7}{2}$ | M1 | |
| | $m_{\text{new}} = \frac{5}{2}$ | | |
| | $y = \frac{5x}{2} + 4$ | A1 | |
| |   | | |
| 5 a) | $\cos A = -\frac{15}{17}, \quad \sin B = -\frac{8}{17}$ | M1 M1 | 5 m |
| | $\sin 2A = 2\sin A\cos A = -\frac{240}{289}$ | A1 | |
| | $\cos(A+B) = \cos A\cos B - \sin A\sin B = -\frac{161}{289}$ | A1 | |

| NO | SOLUTION | MARK |
|---|---|---|
| 5 b) | $90^\circ \times \frac{\pi}{180} = \frac{\pi}{2}$ M1 A1<br>$\frac{\pi}{6} \times \frac{180}{\pi} = 30^\circ$ A1 | 3 m |
| 5 c) | $14^2 + 48^2 = 50^2$ M1<br>$\tan A = \frac{14}{48} \rightarrow A = 16.26^\circ$ A1<br>$B = 90^\circ - A = 73.74^\circ$ A1<br>  | 5 m |